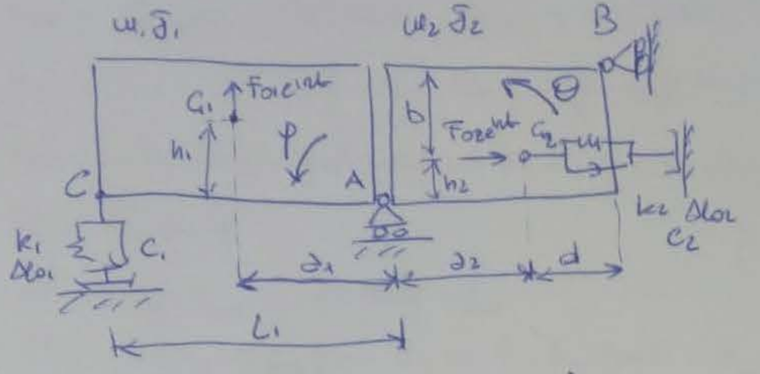
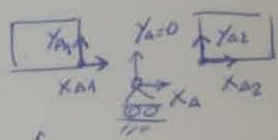
WRITTEN TEST - January 11th 2022

PROBLEM 1



- 2 rigid bodies : 6 dof
- 1 cart B : -1 dof
- cart A :



$$\left\{\begin{array}{l} Y_{A1}=Y_A=0 \\ Y_{A2}=Y_A=0 \\ X_{A1}=X_{A2}=X_A \end{array}\right. \rightarrow 3 \text{ doc.}$$

2 dof.

$$\underline{x}=\begin{Bmatrix}\theta \\ \varphi\end{Bmatrix} \quad ; \quad \underline{x}_0=\begin{Bmatrix}\emptyset \\ \emptyset\end{Bmatrix}$$

$$T=\frac{1}{2}m_1\dot{x}_{G1}^2+\frac{1}{2}m_1\dot{y}_{G1}^2+\frac{1}{2}J_1\dot{\varphi}^2+\frac{1}{2}m_2\dot{x}_{G2}^2+\frac{1}{2}m_2\dot{y}_{G2}^2+\frac{1}{2}J_2\dot{\theta}^2$$

$$D=\frac{1}{2}\sum_{i=1}^{2}c_i\,\Delta\dot{l}_i^2 \quad ; \quad V_{el}=\frac{1}{2}\sum_{i=1}^{2}k_i\,\Delta l_i^2 \quad ; \quad V_g=\sum_{i=1}^{2}m_i g h_{Gi}$$

$$\delta W=\vec{F}_{01}e^{i\Omega t}\times\delta\vec{y}_{G1}+\vec{F}_{02}e^{i\Omega t}\times\delta\vec{x}_{G2}=F_{01}e^{i\Omega t}\,\delta y_{G1}+F_{02}e^{i\Omega t}\,\delta x_{G2}$$

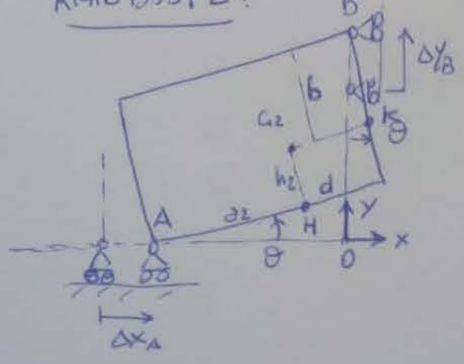
$$[m_{PH}]=\text{diag}(m_1\ m_1\ J_1\ m_2\ m_2\ J_2) \quad ; \quad \underline{y}_m=\{x_{G1}\ y_{G1}\ \varphi\ x_{G2}\ y_{G2}\ \theta\}^T$$

$$[c_{PH}]=\text{diag}(c_1\ c_2) \quad ; \quad [k_{PH}]=\text{diag}(k_1\ k_2) \quad ; \quad \underline{y}_c=\underline{y}_k=\{\Delta l_1\ \Delta l_2\}^T$$

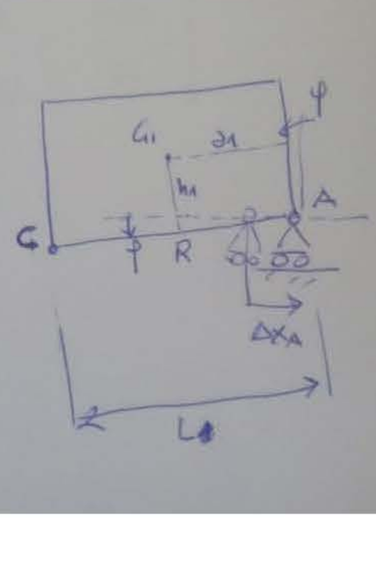
$$\underline{F}=\{F_{01}\ F_{02}\}^T e^{i\Omega t} \quad ; \quad \underline{y}_e=\{y_{G1}\ x_{G2}\}^T$$

Non-linear kinematic relationships :

RIGID BODY 2 :



RIGID BODY 1 :



$$X_{G2} = \bar{BK}\,\text{sen}\,\theta - \bar{G_2K}\cos\theta = b\,\text{sen}\,\theta - d\cos\theta$$

$$Y_{G2} = \bar{AH}\,\text{sen}\,\theta + h_2\cos\theta = a_2\,\text{sen}\,\theta + h_2\cos\theta$$

$$X_A = (\bar{BK} + \bar{G_2H})\,\text{sen}\,\theta - (\bar{G_2K} + \bar{AH})\cos\theta = (b+h_2)\,\text{sen}\,\theta - (d+a_2)\cos\theta$$

$$X_{G1} = X_A - \bar{AR}\cos\varphi - \bar{G_1R}\,\text{sen}\,\varphi = (b+h_2)\,\text{sen}\,\theta - (d+a_2)\cos\theta - a_1\cos\varphi - h_1\,\text{sen}\,\varphi$$

$$Y_{G1} = -\bar{AR}\,\text{sen}\,\varphi + \bar{G_1R}\cos\varphi = -a_1\,\text{sen}\,\varphi + h_1\cos\varphi$$

$$Y_C = -L\,\text{sen}\,\varphi$$

$$\Delta L_1 = Y_C - Y_{C\phi} = -L\,\text{sen}\,\varphi - Y_{C\phi}$$

$$\Delta L_2 = -(X_{G2} - X_{G2\phi}) = -(b\,\text{sen}\,\theta - d\cos\theta) + X_{G2\phi}$$

$$\frac{\partial X_{G2}}{\partial\theta} = b\cos\theta + d\,\text{sen}\,\theta \quad ; \quad \frac{\partial X_{G2}}{\partial\varphi} = 0$$

$$\frac{\partial Y_{G2}}{\partial\theta} = a_2\cos\theta - h_2\,\text{sen}\,\theta \quad ; \quad \frac{\partial Y_{G2}}{\partial\varphi} = 0$$

$$\frac{\partial X_{G1}}{\partial\theta} = (b+h_2)\cos\theta + (d+a_2)\,\text{sen}\,\theta \quad ; \quad \frac{\partial X_{G1}}{\partial\varphi} = a_1\,\text{sen}\,\varphi - h_1\cos\varphi$$

$$\frac{\partial Y_{G1}}{\partial\theta} = 0 \quad ; \quad \frac{\partial Y_{G1}}{\partial\varphi} = -a_1\cos\varphi - h_1\,\text{sen}\,\varphi$$

$$\frac{\partial \Delta L_1}{\partial\theta} = 0 \quad ; \quad \frac{\partial \Delta L_1}{\partial\varphi} = -L\cos\varphi$$

$$\frac{\partial \Delta L_2}{\partial\theta} = -\frac{\partial X_{G2}}{\partial\theta} \quad ; \quad \frac{\partial \Delta L_2}{\partial\varphi} = -\frac{\partial X_{G2}}{\partial\varphi}$$

$$[\Lambda_m] = \begin{bmatrix} b+h_2 & -h_1 \\ 0 & -a_1 \\ 0 & 1 \\ b & 0 \\ a_2 & 0 \\ 1 & 0 \end{bmatrix} \qquad [\Lambda_k] = [\Lambda_c] = \begin{bmatrix} 0 & -L \\ -b & 0 \end{bmatrix} \qquad [\Lambda_Q] = \begin{bmatrix} 0 & -a_1 \\ b & 0 \end{bmatrix}$$

$$\begin{cases} h_{G1} = Y_{G1} \\ h_{G2} = Y_{G2} \end{cases}$$

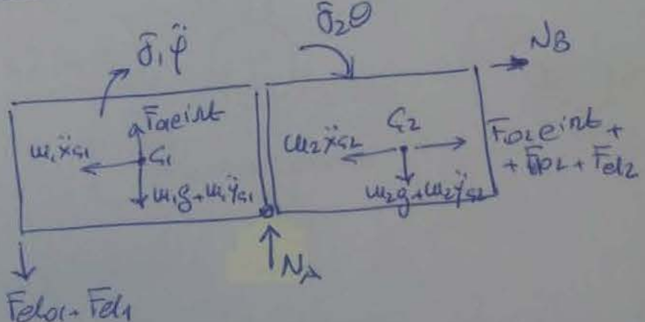
$$\begin{cases} \dfrac{\partial^2 h_{G1}}{\partial \theta^2} = \dfrac{\partial^2 h_{G1}}{\partial \theta \partial \varphi} = 0 \\ \dfrac{\partial^2 h_{G1}}{\partial \varphi^2} = a_1 \operatorname{sen}\varphi - h_1 \cos\varphi \end{cases} \rightarrow [k_{G1}] = m_1 g \begin{bmatrix} 0 & 0 \\ 0 & -h_1 \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial^2 h_{G2}}{\partial \theta^2} = -a_2 \operatorname{sen}\theta - h_2 \cos\theta \\ \dfrac{\partial^2 h_{G2}}{\partial \varphi^2} = \dfrac{\partial^2 h_{G2}}{\partial \theta \partial \varphi} = 0 \end{cases} \rightarrow k_{G2} = m_2 g \begin{bmatrix} -h_2 & 0 \\ 0 & 0 \end{bmatrix}$$

$$\begin{cases} \dfrac{\partial^2 \Delta L_1}{\partial \theta^2} = \dfrac{\partial^2 \Delta L_1}{\partial \theta \partial \varphi} = 0 \\ \dfrac{\partial^2 \Delta L_1}{\partial \varphi^2} = L \operatorname{sen}\varphi \end{cases} \rightarrow [k_{el\,II\,1}] = [0]$$

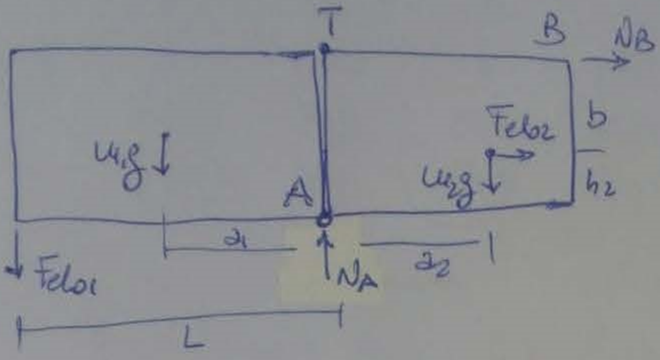
$$\begin{cases} \dfrac{\partial^2 \Delta L_2}{\partial \theta^2} = b \operatorname{sen}\theta - d\cos\theta \\ \dfrac{\partial^2 \Delta L_2}{\partial \varphi^2} = \dfrac{\partial^2 \Delta L_2}{\partial \theta \partial \varphi} = 0 \end{cases} \rightarrow [k_{el\,II,2}] = k_2 \Delta l_{02} \begin{bmatrix} -d & 0 \\ 0 & 0 \end{bmatrix}$$

- question 4 :



$$\Sigma F_x = 0 : \quad N_B = m_1 \ddot{x}_{G1} + m_2 \ddot{x}_{G2} \underbrace{- F_{el02}}_{\text{static}} - F_{el2} - F_{02} e^{i\Omega t}$$

$$\Sigma F_y = 0 : \quad N_A = \underbrace{(m_1 g + m_2 g + F_{el01})}_{\text{static}} + m_1 \ddot{y}_{G1} + m_2 \ddot{y}_{G2} + F_{el1} - F_{01} e^{i\Omega t}$$



$$\sum M_A^{BODY1} = 0 :$$

$$F_{01} L + m_1 g a_1 = 0$$

$$\rightarrow \Delta l_{01} = -\frac{m_1 g}{k_1} \frac{a_1}{L_1}$$

$$\begin{cases} \sum F_x = 0 \rightarrow N_B + F_{el02} = 0 \\ \sum M_A^{BODY2} = 0 \rightarrow N_B(b+h_2) + F_{el02} h_2 + m_2 g a_2 = 0 \end{cases}$$

$$\begin{bmatrix} b+h_2 & h_2 \\ 1 & 1 \end{bmatrix} \begin{Bmatrix} N_B \\ F_{el02} \end{Bmatrix} = \begin{Bmatrix} -m_2 g a_2 \\ 0 \end{Bmatrix} \rightarrow \Delta l_{02} = \frac{F_{el02}}{k_2}$$

or: $\sum M_T^{BODY2} = 0 \quad : \quad F_{el02}\, b - m_2 g a_2 = 0 \rightarrow \Delta l_{02} = \frac{m_2 g}{k_2} \frac{a_2}{b}$